



Practical - 30

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30) Cluster a dataset into two groups using k-means clustering in Breeze.

Code:

```
import breeze.linalg._
```

```
object prac30 extends App {
```

```
  // Dataset
```

```
  val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
```

```
  // Initial cluster centers
```

```
  var center1 = 2.0
```

```
  var center2 = 10.0
```

```
  // Repeat clustering
```

```
  for (i <- 1 to 5) {
```

```
    // Assign points to Cluster 1
```

```
    val cluster1 = data.toArray.filter(x =>
```

```

    math.abs(x - center1) < math.abs(x - center2)
  )

  // Assign points to Cluster 2
  val cluster2 = data.toArray.filter(x =>
    math.abs(x - center2) <= math.abs(x - center1)
  )

  // Calculate new centers
  center1 = cluster1.sum / cluster1.length
  center2 = cluster2.sum / cluster2.length
}

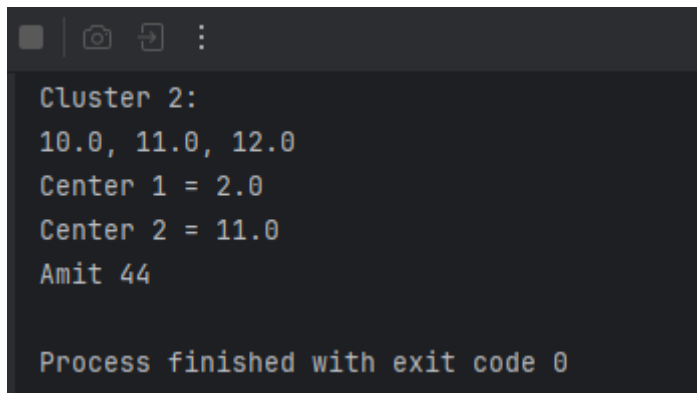
// Display clusters
println("Cluster 1:")
println(
  data.toArray
    .filter(x => math.abs(x - center1) < math.abs(x - center2))
    .mkString(", ")
)

println("Cluster 2:")
println(
  data.toArray
    .filter(x => math.abs(x - center2) <= math.abs(x - center1))
    .mkString(", ")
)

println("Center 1 = " + center1)
println("Center 2 = " + center2)
}

```

Output:

A terminal window with a dark background and light gray text. The window has a title bar with standard OS icons (close, maximize, minimize, and a menu icon). The output text is as follows:

```
Cluster 2:  
10.0, 11.0, 12.0  
Center 1 = 2.0  
Center 2 = 11.0  
Amit 44  
  
Process finished with exit code 0
```